

High Level Flow: Data Pipeline

Source (CSV / S3)

↓ Read from raw layer

Data Cleaning & Validation

↓ Null handling, deduplication, filtering invalid records & date conversion

Ingestion to staging layer

↓ storing the financial data for historical references

Data Enrichment

↓ Apply business rules (risk-calculation with time based risk)

Aggregations

↓ Analytical metrics time based & Reporting

Storage (Parquet)

↓ storing in Data Lake for dash boards/reporting tools/analytical purposes

Load into Redshift

↓ for dashboards & BI tools

Monitoring & Logging

↓

CI/CD Automation

↓ push the code to repository

Orchestrate Scheduled to run using Streamsets/Airflow DAG pipeline

Tools & Technologies:

PySpark	Scalable/parallel data processing
S3	Durable/Reliable/Budget friendly storage
Parquet	Columnar as its compressed, saving space and faster IO
Bitbucket/Git	Simple CI/CD
Logging info in ETL.py	Monitoring/Notifications

CI/CD

- Git-based workflow
- Automated testing & deployment

Data Quality & Testing

- Deduplication
- Null & range checks
- Schema validation
- Row-count reconciliation

Monitoring & Operations

- Logging
- Data volume tracking
- Failure alerts

Challenges & Solutions

Data quality issues	Validation layer
Scaling	Spark + Parquet
Failures	Idempotent writes

Scalability & Agility (Finance Context)

- Handles **high transaction volumes** because of scalability processing using pyspark & **storage in S3**
- Supports **regulatory audits**
- Easy onboarding of new data sources